NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2024-2025

EE2005 – ELECTRICAL DEVICES AND MACHINES

November / December 2024 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. (a) Consider the magnetic circuit shown in Figure 1, where a N-turn coil with resistance of 2.5 Ω is wound on the center limb. The mean lengths of the air gap and the various sections of the core are as indicated in the figure, where l_g = 2 mm, $l_1 = l_4 = 49.9$ cm, l_2 = 20 cm, and l_3 = 100 cm. The cross-sectional area of each limb is 36 cm^2. Assume a relative permeability of the core to be 2500. The permeability of free space is $\mu_o = 4\pi x10^{-7}$ H/m. You may ignore any leakage and fringing.

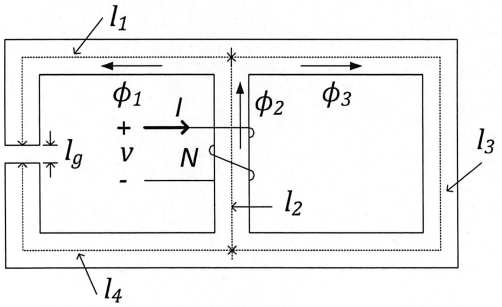


Figure 1

Note: Question No. 1 continues page 2.

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(i) Draw the equivalent circuit of the given magnetic circuit and indicate clearly the values of the reluctance of each section of the magnetic circuit. Determine the fluxes ϕ_1, ϕ_2 and ϕ_3 (in mWb) if the current I is 4 A and N is 100.

(ii) Determine the energies stored in the air gap, in the core of the left limb (excluding the air gap), in the center limb and the right limb. Also, determine the inductance of the coil.

(iii) Without having to repeat the calculations like in part (i), determine the new fluxes ϕ_1, ϕ_2 and ϕ_3 (in mWb) if the current I is now reduced to 2 A.

(19 Marks)

(b) Suppose now the 100-turn coil is connected to a 50-Hz AC supply. Calculate the amplitudes of the flux ϕ_2, the emf induced in the coil and the supply voltage v if the amplitude of the current I in the coil is 2.5 A.

(6 Marks)

2. (a) The approximate equivalent circuit of the transformer can be determined by conducting two common tests. The two common tests were conducted on a 23-kVA, 2300-V/230-V, 50-Hz step-down transformer with the following readings:

Test 1: Voltage = 230 V, Current = 2 A, Power = 60 W

Test 2: Voltage = 120 V, Current = 10 A, Power = 400 W

(i) What tests do Test 1 and Test 2 refer to? Justify your answers. Hence, using these test data, find the approximate equivalent circuit of the transformer referred to the secondary side, indicating the parameter values.

(ii) Using the approximate equivalent circuit in part (i), calculate the efficiency of the transformer at 50% rated load, 0.85 power factor lagging and rated secondary voltage. Use the rated secondary voltage as the reference, that is, $\vec{V}_2 = V_2 \angle 0^o$.

(17 Marks)

(b) The two-winding transformer in part (a) is now used as an autotransformer to supply 2530 V from a 2300 V supply.

(i) Draw the specified autotransformer connection and calculate its KVA rating.

(ii) Determine the efficiency of the autotransformer at 50% rated load and 0.85 power factor lagging. State any assumption(s) made.

(8 Marks)

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3. (a) A three-phase, 440-V, 50-Hz, 6-pole, wye-connected induction motor draws a line current of 50 A at a lagging power factor of 0.85 from the supply when it is running at a speed of 950 rpm. The rotational losses of the motor are 1500 W and the stator resistance per phase is $0.08\ \Omega$. Determine

(i) The rotor copper losses.

(ii) The load torque of the motor.

(iii) The efficiency of the motor.

(6 Marks)

(b) A three-phase, 3300-V, 50-Hz, 6-pole, wye-connected induction motor has the following per phase parameters referred to the stator:

$$R_1 = 3\ \Omega,\ R_2 = 3\ \Omega,\ X_M = 350\ \Omega,\ X_1 = 5\ \Omega,\ X_2 = 5\ \Omega$$

Determine

(i) The maximum torque developed by the motor and the slip at which it occurs.

(ii) The magnitude of the input current drawn by the motor from the supply and the efficiency of the motor if the motor is running at a slip of 5 %. The rotational losses are negligible.

(13 Marks)

(c) Consider a three-phase, 50-Hz, 4-pole, wye-connected inductor which has negligible stator impedance, i.e., $R_1 = 0\ \Omega$ and $X_1 = 0\ \Omega$, and with $R_2 = 4.5\ \Omega$, $X_M = 100\ \Omega$ and $X_2 = 8.5\ \Omega$. Let R_{ext} be the external resistance that is added in each phase of the rotor circuit and referred to the stator. If $R_{ext} = 0\ \Omega$, the starting torque developed by the motor is 85 Nm. Determine the starting torque developed by the motor if $R_{ext} = 3\ \Omega$.

(6 Marks)

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4. (a) A dc shunt motor has an armature winding resistance and a field winding resistance of 0.8 Ω and 150 Ω, respectively. The armature terminals are connected to a 240-V dc supply.

(i) When the motor is operating at a full-load speed of 2000 rpm, the power input to the motor is 4800 W. Determine the torque developed by the motor at full load.

(ii) If the no-load speed of the motor is 2100 rpm, determine the no-load armature current and the rotational losses of the motor. Determine also the efficiency of the motor at full load.

(iii) An external resistance $R_{ext} = 3.2\ \Omega$ is now connected in series with the armature winding and the torque developed by the motor is the same as the torque developed at full load. Determine the speed of the motor under this condition.

(15 Marks)

(b) A separately-excited dc generator has an armature winding resistance of 0.6 Ω, a field winding resistance of 120 Ω and constant rotational losses of 300 W. Neglect armature reaction and assume that the speed of the generator's external prime mover is constant.

(i) Suppose that the field winding is connected to a 120-V dc supply and at no load, the internal generated voltage is 200 V. Determine the efficiency of the generator when a load, represented by an equivalent resistance $R_L = 5\ \Omega$, is connected to the armature terminals.

(ii) Consider now that an external variable resistance R_{ext} is connected in series with the field winding and the field circuit is fed by a 120-V supply. Determine the value of R_{ext} required if the generator is delivering a current of 25 A to the load of $R_L = 5\ \Omega$. Assume that the machine is operating in the linear part of its magnetization curve.

(10 Marks)

END OF PAPER

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1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.

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